

Comprehensive Report: Phase-Change Kinetics and Ballistic Modification of Fuel-Rich KNSB Propellant (Batch 03)

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1. Introduction & Ballistic Objectives

Standard potassium nitrate and sorbitol (KNSB) formulations traditionally utilize a stoichiometric 65:35 oxidizer-to-fuel ratio to maximize specific impulse I_{sp} and characteristic velocity. However, for specific mission profiles requiring lower chamber pressures and managed structural regression rates, the thrust profile must be systematically suppressed. Batch 03 was formulated as an experimental deviation from the baseline, introducing a fuel-rich 60:40 mass ratio. The structural and ballistic objectives of this alteration are twofold:

- Thrust Profile Suppression:** To reduce the mass burn rate and core pressure peak without inducing critical combustion instability.
- Mechanical Property Preservation:** To leverage the excess sorbitol as an internal plasticizing agent, thereby decreasing matrix brittleness and preventing micro-fracturing along the grain core during ignition transient phases.

2. Experimental Setup and Material Specifications

The processing environment was isolated to maintain uniform ambient thermal conditions. The batch preparation utilized commercial-grade liquid Sorbitol 70 and granular Potassium Nitrate KNO_3 , milled to a target particle size distribution to ensure rapid wet-slurry incorporation.

- Initial Binder Charge:** 866.0 g of liquid Sorbitol 70.
- Theoretical Anhydrous Target:** Calculated at 606.0 g based on an estimated 30% aqueous phase reduction.
- Structural Containment:** A lightweight, pre-tared protective liner casing measuring exactly 100.0 g.
- Oxidizer Charge:** 929.0 g of fine-grain KNO_3

3. Dehydration Kinetics & Thermal Step Logs

The dehydration phase commenced precisely at 10:13 AM. The mixture was processed on an adjustable electric heating element, with an initial baseline thermal setpoint established at 120°C. Mass indicators and core temperatures were logged dynamically at strict 10-minute intervals to evaluate the moisture extraction profile.

Table I: Chronological Dehydration and Thermal Matrix Logs

Timestamp	Elapsed Time (min)	Core Temp (°C)	Current Fuel Mass (g)	Incremental Delta (g)	Cumulative Loss (g)
10:13 AM	0 min	120	866	—	0

10:23 AM	10 min	120	837	-29	29
10:33 AM	20 min	120	710	-127	156
10:43 AM	30 min	120	706	-4	160
10:53 AM	40 min	120	655	-51	211
11:03 AM	50 min	120	640	-15	226
11:13 AM	60 min	120	624	-16	242
11:23 AM	70 min	90	622	-2	244
11:33 AM	80 min	90	619	-3	247

3.1 Kinetic Evaporation Curve Analysis

The moisture extraction rate exhibited non-linear progression, highly characteristic of concentrated polyol solutions.

- **Phase I (0–20 Minutes):** A rapid, highly volatile boiling phase occurred, peaking between 10:23 AM and 10:33 AM with a major mass drop of 127.0 g. This indicates the rapid liberation of unbound, free water from the slurry matrix.
- **Phase II (20–60 Minutes):** The evaporation curve constricted significantly. The minor 4.0 g loss between 20 and 30 minutes followed by a secondary 51.0 g plunge suggests structural water-binding and pocket-steam venting within the increasingly viscous fluid.
- **Phase III (60–80 Minutes):** At 11:23 AM (70 minutes elapsed), the processing temperature was stepped down from 120°C to 90°C. This tactical adjustment was executed to protect the fuel-rich matrix from accidental caramelization or thermal degradation as the moisture mass thinned out.

3.2 Terminal Operational Saturation Point

Ideally, based on standard molecular hydration curves, the fuel mass should have driven down to an absolute anhydrous limit of 606.0 g. However, between 11:13 AM and 11:33 AM, the mass delta converged flatly from 624.0 g to 622.0 g, and finally stabilized at 619.0 g.

With an incremental change of only 3.0 g over the final 10 minutes at a stabilized temperature, the processing team concluded that the binder had reached its true operational saturation point under current laboratory ambient humidity limits. Forcing further water loss would risk scorching the sugars; thus, 619.0 g was locked as the active dry fuel mass.

4. Stoichiometric Alteration & Blending Mechanics

To lock down the intended 60:40 configuration using the real-world operational saturation fuel mass 619.0 g, the absolute dry oxidizer mass was calculated using the following state relation:

$$\text{KNO} = (619.0 \text{ g} \times 0.40) \times 0.60 = 928.5 \text{ g}$$

To adapt to the 1.0 g physical resolution threshold of the laboratory weighing balance, exactly 929.0 g of milled KN was weighed out.

The oxidizer was integrated incrementally into the molten sorbitol fluid at 90°C. Because of the higher volume fraction of fuel (40% vs the nominal 35%), the slurry demonstrated excellent fluidity and mechanical wetting, resulting in highly uniform crystalline dispersion during the mechanical agitation phase.

5. Casting and Post-Cure Summary

The compound was systematically worked to remove trapped air pockets and poured smoothly at a cast temperature of ~60°C into the pre-weighed 100.0 g protective casing liner. The increased fuel ratio successfully prevented the high-viscosity "clumping" typical of standard batches, yielding a homogeneous, dense grain configuration upon solidification. Static test firing protocols will monitor this specific batch to determine the exact burn rate exponent variance caused by this fuel-rich shift.